Course Title: Ethical Hacking

Objective

The objective of this experiment is to perform reconnaissance and system monitoring using Sysmon, theHarvester, WHOIS, and Netdiscover. These tools help collect information about systems, domains, and networks, monitor system activities, identify active devices, and gather publicly available information for cybersecurity analysis and incident investigation.

Methodology

1. Install and configure Sysmon to monitor system activities and generate security event logs.
2. Use theHarvester to gather publicly available information such as email addresses, subdomains, hostnames, and IP addresses related to the target domain.
3. Perform WHOIS queries to retrieve domain registration details, registrar information, nameservers, and registration dates.
4. Use Netdiscover to scan the local network and identify active hosts, IP addresses, MAC addresses, and device vendors.
5. Analyze the information collected from all tools to understand the system, network, and domain environment and identify potential security risks.

Expected Outcome

By completing this experiment, you will be able to:

- Monitor system activities and generate detailed security logs using Sysmon.
- Collect publicly available information about target domains using theHarvester.
- Retrieve domain registration and ownership details using WHOIS.
- Discover active devices on a local network using Netdiscover.
- Gain practical experience with reconnaissance and system monitoring techniques.
- Improve the ability to analyze systems and networks for cybersecurity monitoring, incident response, and security assessments.

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1.Introduction to Sysmon (System Monitor)

Sysmon (System Monitor) is a free Windows system monitoring tool developed by Microsoft. It runs as a Windows service and records detailed system activity in the Windows Event Log, helping administrators and cybersecurity professionals detect suspicious or malicious behavior.

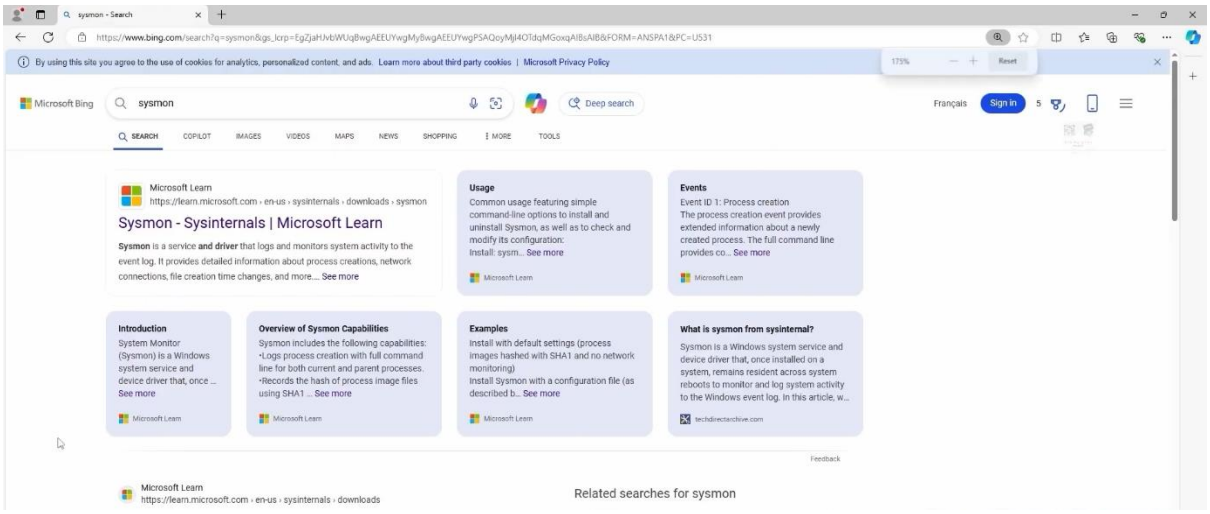
1.1. Uses of Sysmon

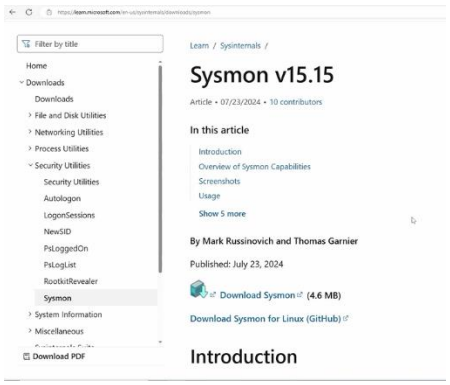
- Monitors process creation and termination.
- Records file creation time changes.
- Monitors network connections.
- Detects driver and DLL loading.
- Records registry modifications.
- Monitors file creation and deletion

2. Setting up Sysmon

Step 1: Download Sysmon

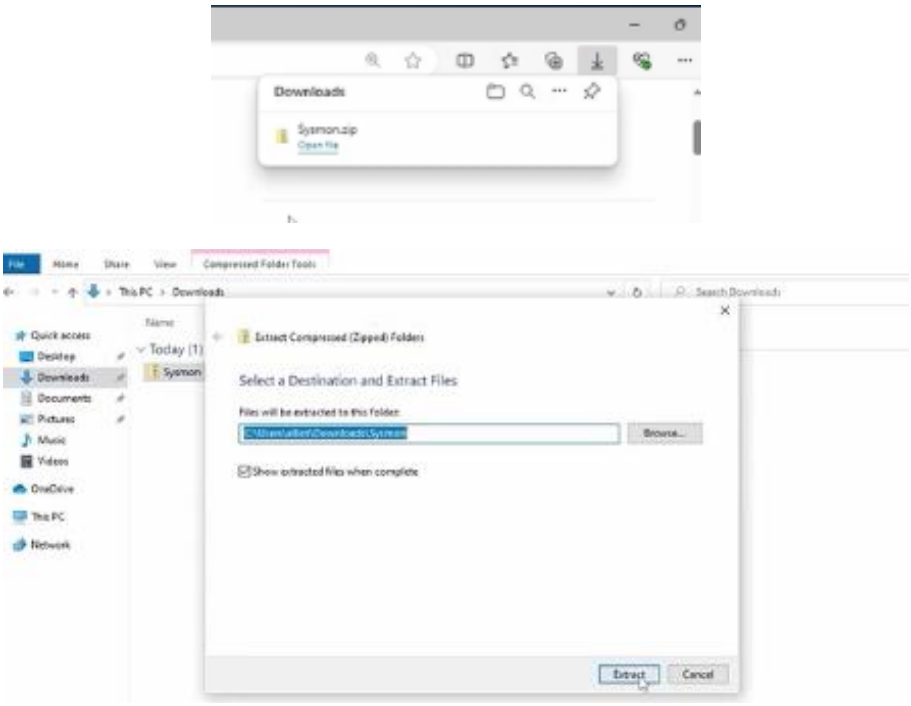
Search for Sysmon and download Sysmon from the Sysinternals Suite.





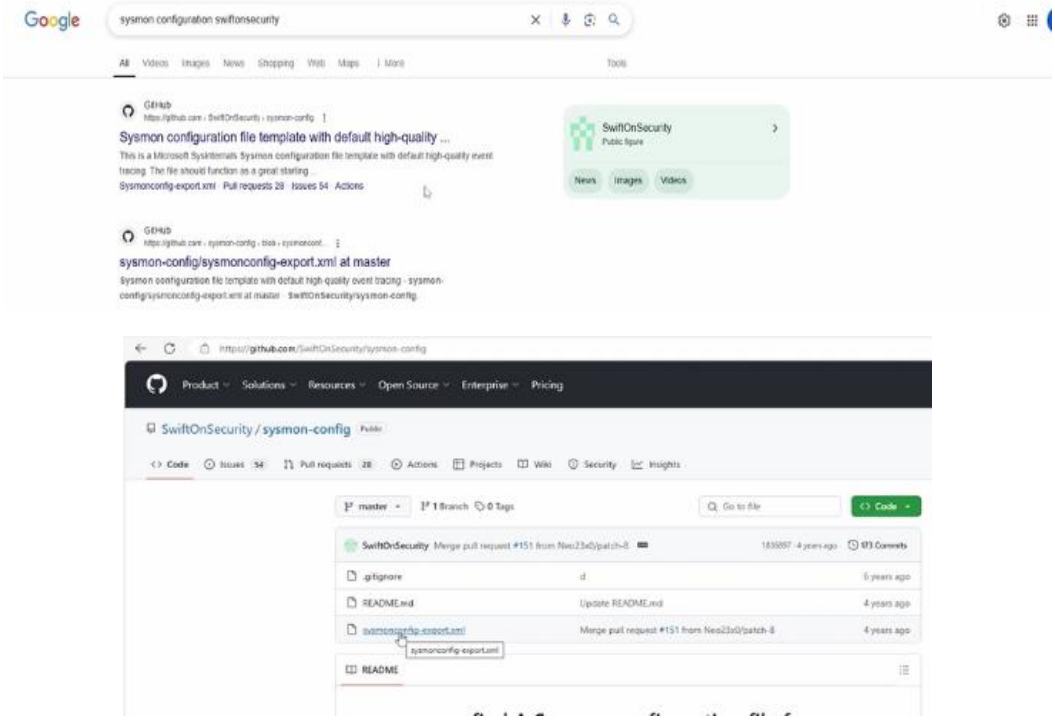
Step 2: Extract the Zip file

Extract the ZIP file from the downloaded file.



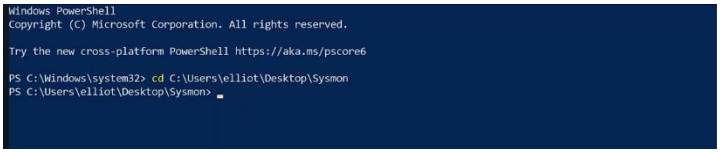
Step 3: Download configuration file

This installs Sysmon with custom monitoring rules defined in sysmonconfig.xml.



Step 4: Open Command Prompt

Open Command Prompt or PowerShell as Administrator.



Step 5: Install Sysmon

sysmon.exe -i

This installs the Sysmon service using the default configuration.

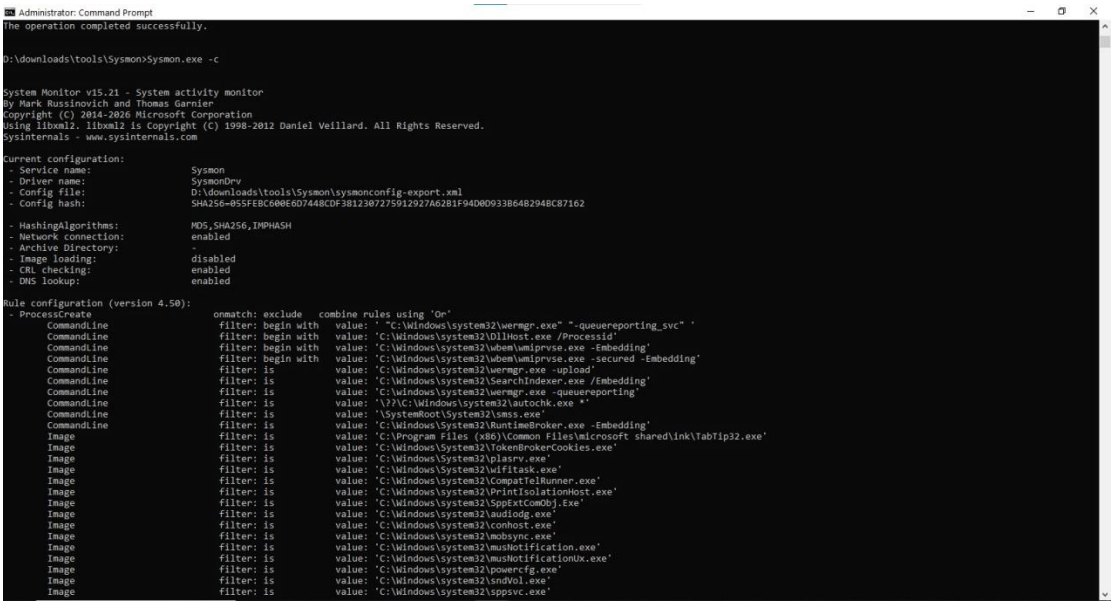


3. Sysmon Commands

3.1. Update the Configuration

sysmon.exe -c sysmonconfig.xml

Updates the current monitoring rules without reinstalling Sysmon.



3.2. View Current Configuration

sysmon.exe -c

Displays the active Sysmon configuration.

3.3. Check Sysmon Version

sysmon.exe -? or sysmon.exe –version

Shows the installed version and available command-line options.

3.4. Uninstall Without Confirmation

sysmon.exe -u force

Removes Sysmon immediately without asking for confirmation.

3.5. Help Command

sysmon -h

-h or --help displays the Sysmon help menu.

```
D:\downloads\tools\Sysmon>Sysmon --help

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Usage:
Install:                Sysmon.exe -i [<configfile>]
Update configuration:   Sysmon.exe -c [<configfile>]
Install event manifest: Sysmon.exe -m
Print schema:           Sysmon.exe -s
Uninstall:              Sysmon.exe -u [force]
  -c   Update configuration of an installed Sysmon driver or dump the
       current configuration if no other argument is provided. Optionally
       take a configuration file.
  -i   Install service and driver. Optionally take a configuration file.
  -m   Install the event manifest (done on service install as well)).
  -s   Print configuration schema definition of the specified version.
       Specify 'all' to dump all schema versions (default is latest)).
  -u   Uninstall service and driver. Adding force causes uninstall to proceed
       even when some components are not installed.

The service logs events immediately and the driver installs as a boot-start driver to capture activity from early in the boot that the service will write to the event log when it starts.

On Vista and higher, events are stored in "Applications and Services Logs/Microsoft/Windows/Sysmon/Operational". On older systems, events are written to the System event log.

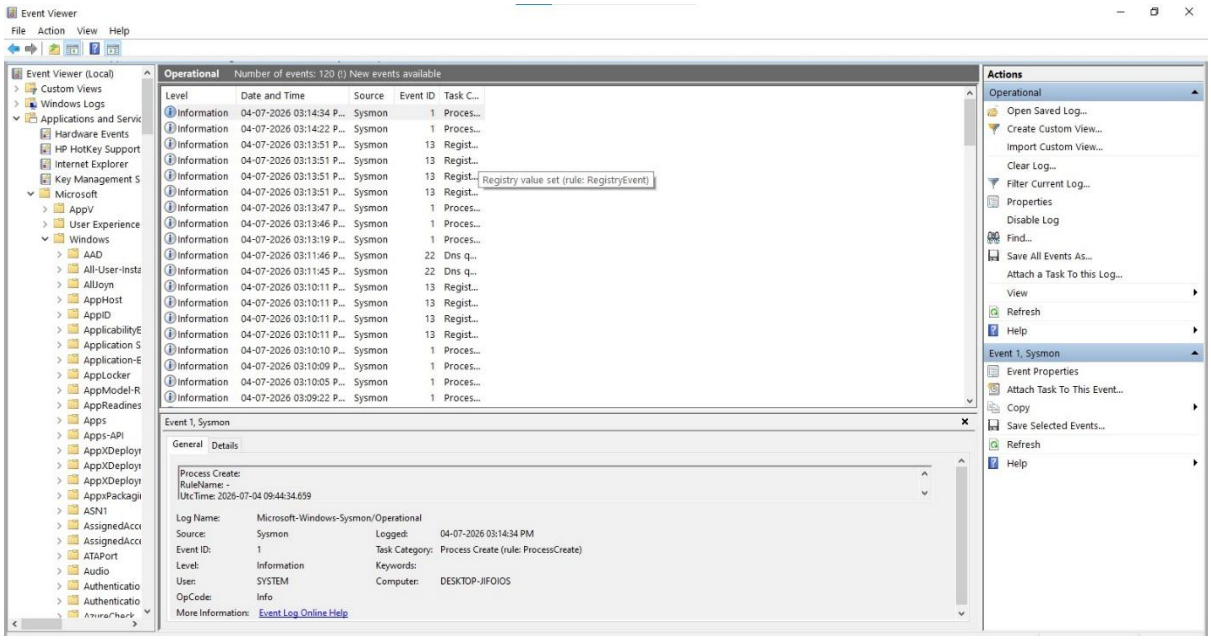
Use the '-? config' command for configuration file documentation. More examples are available on the Sysinternals website.

Specify -accepteula to automatically accept the EULA on installation, otherwise you will be interactively prompted to accept it.

Neither install nor uninstall requires a reboot.
```

Example:

Monitoring system activities in real time.



4. Introduction to theharvester tool

theHarvester is an Open Source Intelligence (OSINT) tool used to collect publicly available information such as email addresses, subdomains, hostnames, IP addresses, and employee names about a target domain. It is commonly used during the reconnaissance phase of cybersecurity assessments.

5. Setting up for theHarvester

5.1. Update the System

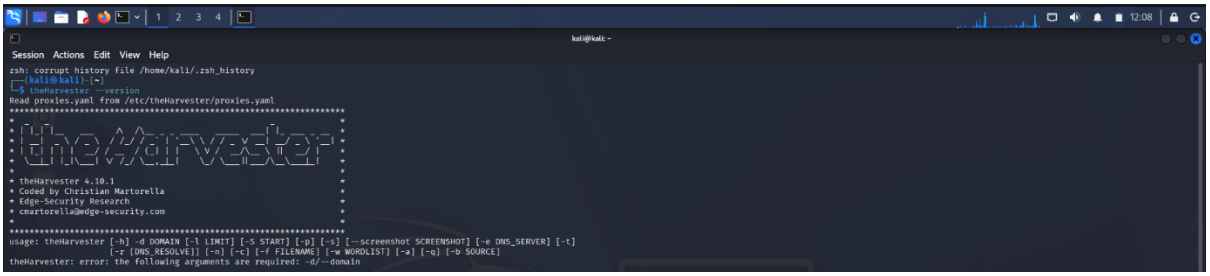
sudo apt update

Updates the package list from the repositories.

5.2. Install theHarvester

sudo apt install theharvester

Installs theHarvester from the distribution's package repository.



5.3. Verify Installation

theHarvester --version

Displays the installed version of theHarvester.

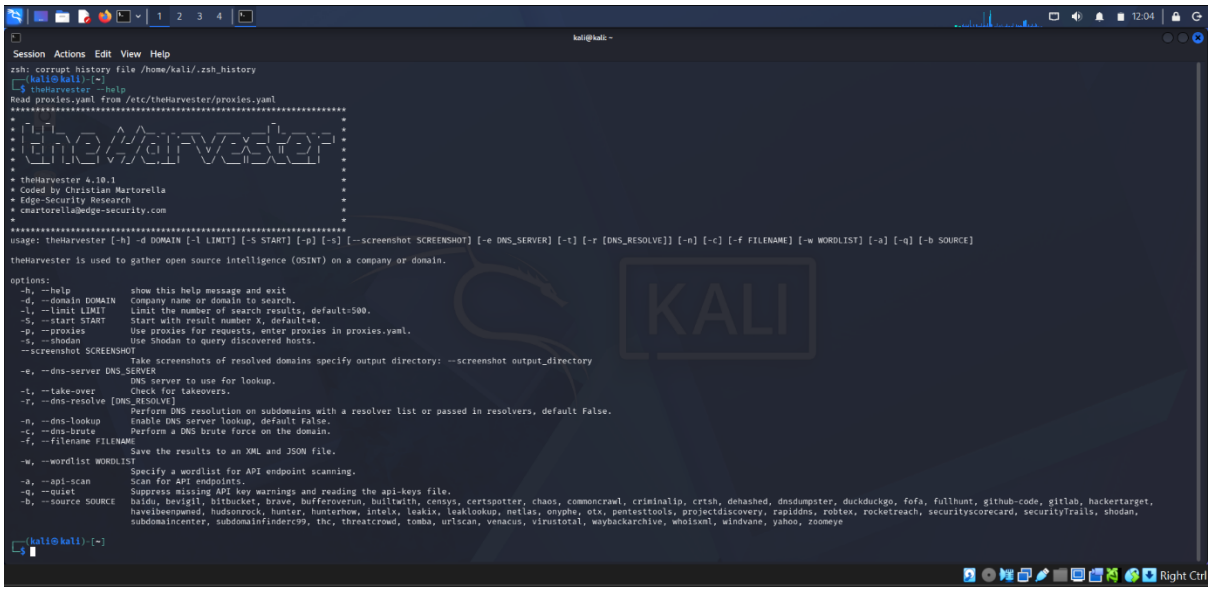
5.4. Display Help

theHarvester -h

or

theHarvester –help

Shows all available options, syntax, and examples.



5.5 Basic Syntax

theHarvester -d <domain> -b <source>

-d → Target domain

-b → Data source to search

Example : theHarvester -d microsoft.com -b all -l 300 –q



6. Introduction to Netdiscover

Netdiscover is a network reconnaissance tool used to discover live hosts on a local network by sending ARP (Address Resolution Protocol) requests. It is commonly used to identify the IP and MAC addresses of devices connected to a LAN.

7. Setting up netdiscover

7.1. Update the package list

sudo apt update

Updates the package repository information.

7.2. Install Netdiscover

sudo apt install netdiscover

Installs Netdiscover on Ubuntu, Debian, or Kali Linux.

```
┌──(riya㉿kali)-[~]
└─$ sudo apt install netdiscover -y
The following package was automatically installed and is no longer required:
  iucode-tool
Use 'sudo apt autoremove' to remove it.

Installing:
  netdiscover

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 393
  Download size: 511 kB
  Space needed: 3,230 kB / 7,837 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 netdiscover amd64 0.21-1 [511 kB]
Fetched 511 kB in 6s (80.5 kB/s)
Selecting previously unselected package netdiscover.
(Reading database… 424759 files and directories currently installed.)
Preparing to unpack …/netdiscover_0.21-1_amd64.deb…
Unpacking netdiscover (0.21-1)…
Setting up netdiscover (0.21-1)…
Processing triggers for kali-menu (2026.2.6)…
Processing triggers for man-db (2.13.1-1)…
```

7.3. Verify Installation

netdiscover --version

Displays the help menu and confirms that Netdiscover is installed.

Basic Syntax

netdiscover [OPTIONS]

8. Commands for netdiscover

8.1. Display Help

netdiscover -h

Shows all available options and command syntax.

8.2. Scan the Default Network Interface

sudo netdiscover

Starts scanning the local network using the default network interface.

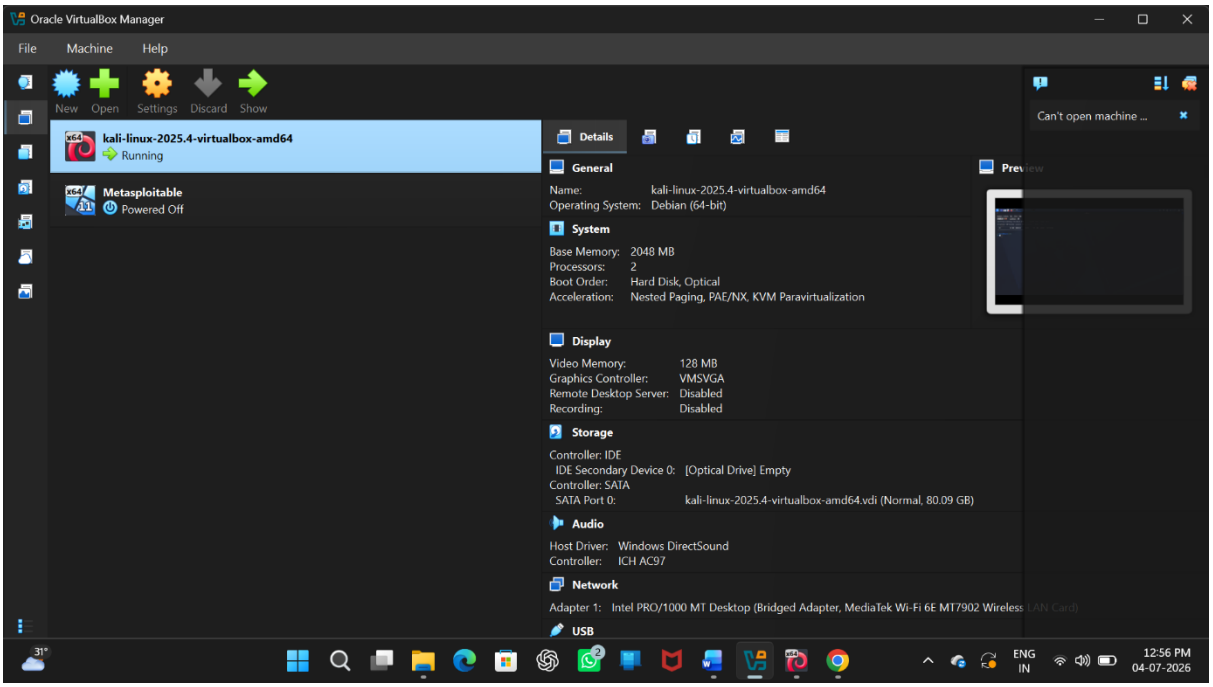
```
Currently scanning: 172.17.0.0/16   |   Screen View: Unique Hosts

1434 Captured ARP Req/Rep packets, from 12 hosts.   Total size: 86040
_____________________________________________________________________________
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
_____________________________________________________________________________
192.168.1.16    34:02:86:9a:b1:90      1      60  Intel Corporate
192.168.1.35    0a:d0:4e:76:c7:d9      1      60  Unknown vendor
192.168.1.2     76:8d:4c:b7:b5:a2      7     420  Unknown vendor
192.168.1.23    4a:24:52:ff:63:4e      1      60  Unknown vendor
192.168.1.9     f2:23:15:f6:ea:aa      1      60  Unknown vendor
192.168.1.24    72:0d:48:3b:bb:5f      1      60  Unknown vendor
192.168.1.12    ea:99:f9:20:9d:d1      1      60  Unknown vendor
192.168.1.27    c2:88:fb:02:c0:3e      2     120  Unknown vendor
192.168.1.32    f0:67:28:a6:13:41      4     240  GUANGDONG OPPO MOBILE TELECOMMUNICATIO
192.168.1.11    7a:56:b7:98:53:ab      1      60  Unknown vendor
192.168.1.33    86:79:8f:82:8a:96      1      60  Unknown vendor
```

8.3. Scan a Specific Network Range

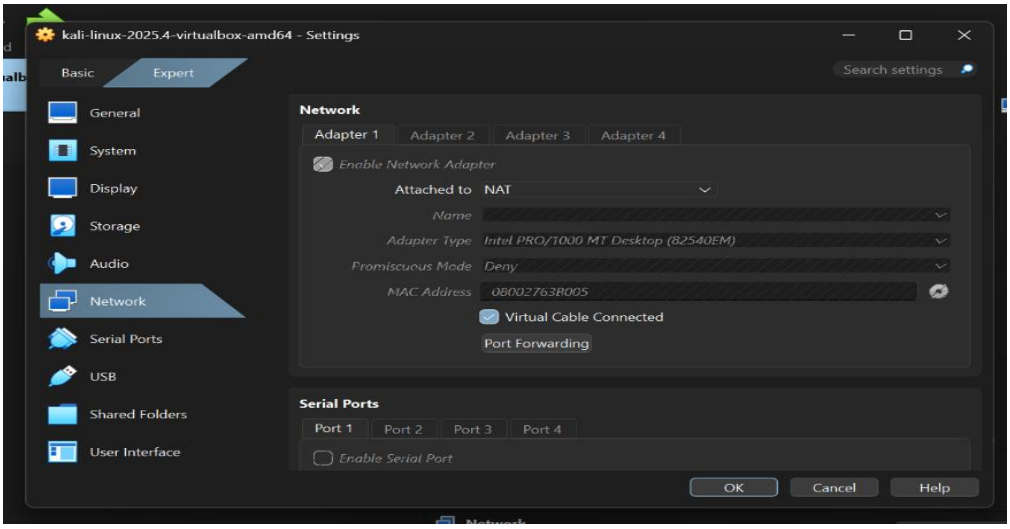
Setting up for netdiscover in the virtualbox

1.Open the settings in the virtual box



2. Enter into network and change NAT to Bridged Adapter in the attached to dropdown option.

sudo netdiscover -r 192.168.1.0/24

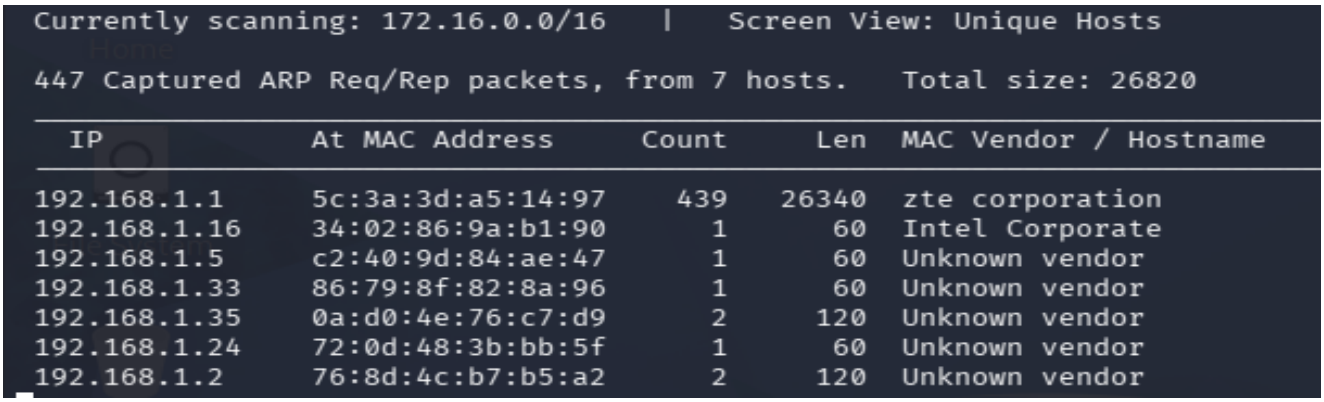


- -r → Specifies the IP range to scan.
- 192.168.1.0/24 → Scans all devices in that subnet.

8.4. Scan Using a Specific Network Interface

sudo netdiscover -i eth0

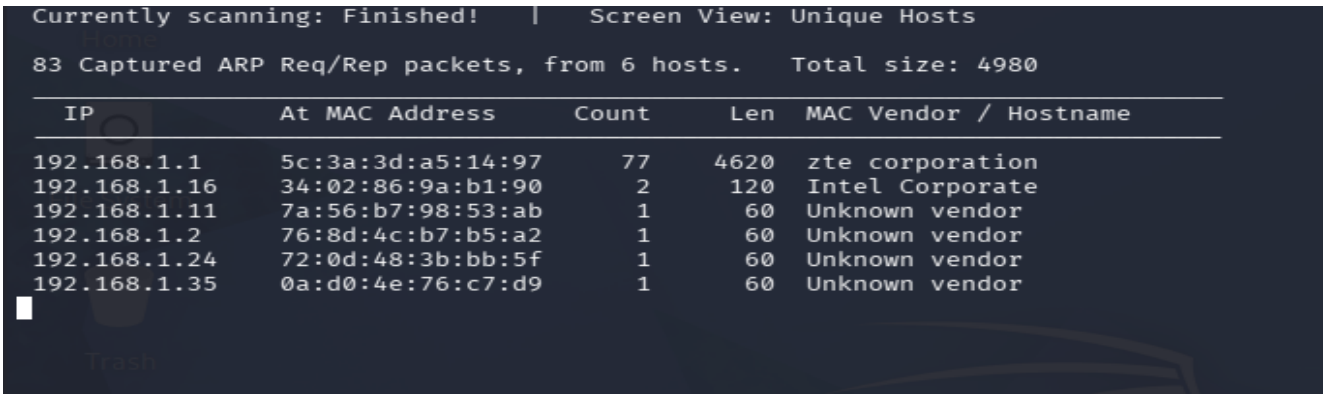
- -i → Specifies the network interface.
- eth0 → Uses the eth0 interface for scanning.



8.5. Scan a Specific Interface and Range

sudo netdiscover -i wlan0 -r 192.168.0.0/24

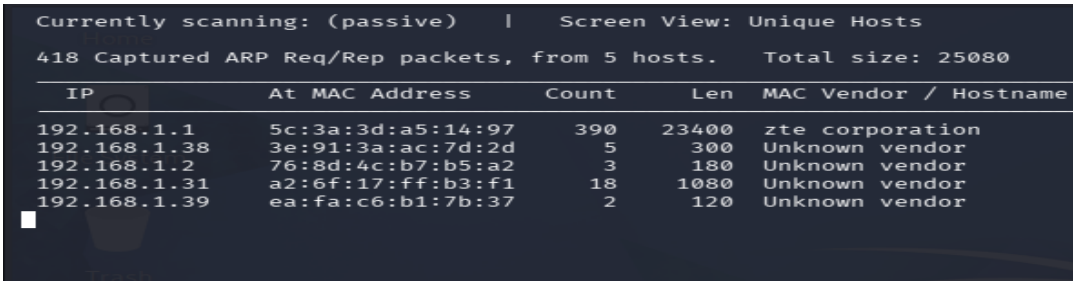
Scans the specified subnet using the wlan0 wireless interface.



8.6. Passive Mode Scan

sudo netdiscover -p

- -p → Enables passive mode.
- Listens for ARP traffic without actively sending ARP requests.



8.7. Ignore Home Network

sudo netdiscover -S

-S → Skips scanning the local/home network and focuses on other detected networks

9. Introduction to WHOIS

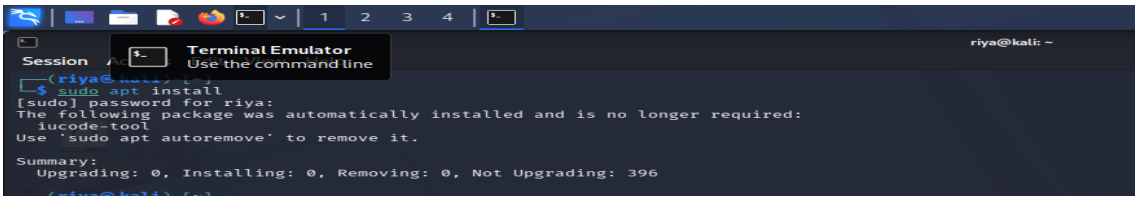
WHOIS is a command-line tool used to retrieve registration information about a domain name or IP address. It provides details such as the registrar, registration dates, nameservers, and registration status.

10. Setup for Whois

10.1. Update the package list

sudo apt update

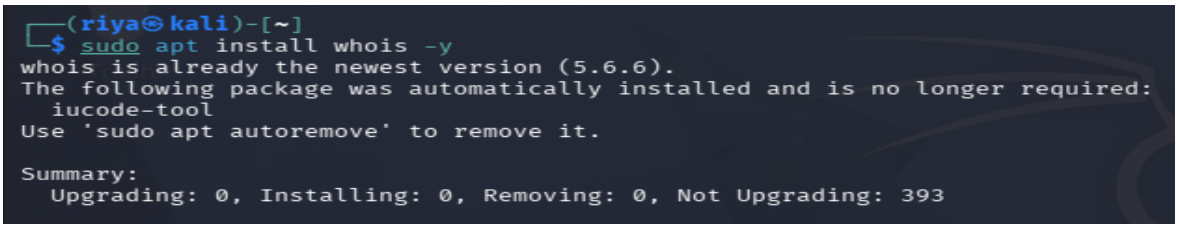
Updates the package repository information.



10.2. Install WHOIS

sudo apt install whois

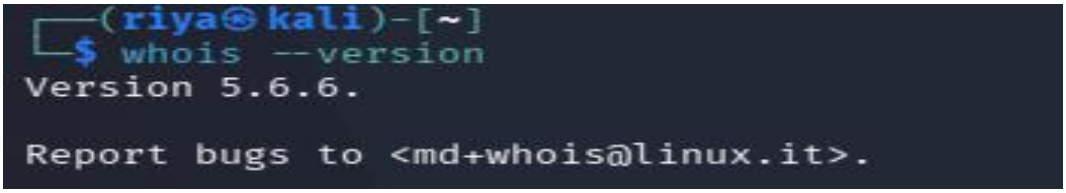
Installs the WHOIS command-line tool.



10.3. Verify Installation

whois –version

Displays the installed version of the WHOIS tool (if supported).



10.4 Basic Syntax

whois <domain_or_ip>

Example:

whois google.com

11. Whois Commands

11.1. Display Help

whois –help

Displays available options and command usage.



11.2. Look Up a Domain

whois example.com

Retrieves registration details for the domain example.com.

```
┌──(riya㉿kali)-[~]
└─$ whois google.com
   Domain Name: GOOGLE.COM
   Registry Domain ID: 2138514_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.markmonitor.com
   Registrar URL: http://www.markmonitor.com
   Updated Date: 2019-09-09T15:39:04Z
   Creation Date: 1997-09-15T04:00:00Z
   Registry Expiry Date: 2028-09-14T04:00:00Z
   Registrar: MarkMonitor Inc.
   Registrar IANA ID: 292
   Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
   Registrar Abuse Contact Phone: +1.2086851750
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: NS1.GOOGLE.COM
   Name Server: NS2.GOOGLE.COM
   Name Server: NS3.GOOGLE.COM
   Name Server: NS4.GOOGLE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-06T15:33:47Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
```

11.3. Look Up an IP Address

whois 8.8.8.8

Displays registration information for the specified IP address.

```
└─$ whois 8.8.8.8

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#


NetRange:       8.8.8.0 - 8.8.8.255
CIDR:           8.8.8.0/24
NetName:        GOGL
NetHandle:      NET-8-8-8-0-2
Parent:         NET8 (NET-8-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Google LLC (GOGL)
RegDate:        2023-12-28
Updated:        2023-12-28
Ref:            https://rdap.arin.net/registry/ip/8.8.8.0


OrgName:        Google LLC
OrgId:          GOGL
Address:        1600 Amphitheatre Parkway
City:           Mountain View
StateProv:      CA
PostalCode:     94043
Country:        US
RegDate:        2000-03-30
Updated:        2019-10-31
Comment:        Please note that the recommended way to file abuse complaints are located in the following links.
Comment:
Comment:        To report abuse and illegal activity: https://www.google.com/contact/
Comment:
Comment:        For legal requests: http://support.google.com/legal
Comment:
Comment:        Regards,
```

11.4. Query a Specific WHOIS Server

whois -h whois.verisign-grs.com example.com

- -h → Specifies the WHOIS server to query.
- Useful when you want to query a particular registry.



12. Introduction to Sublist3r

Sublist3r is an Open Source Intelligence (OSINT) tool used to discover subdomains of a target domain. It collects subdomains from multiple public search engines and online sources, making it useful during the reconnaissance phase of ethical hacking.

13. Setting up Sublist3r

13.1. Update the Package List

sudo apt update

Open the terminal in Kali Linux and update the package repository.

```
┌──(riya㉿kali)-[~]
└─$ sudo apt update
[sudo] password for riya:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.3 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
54% [3 Contents-amd64 13.0 MB/53.5 MB 24%]
81% [3 Contents-amd64 38.2 MB/53.5 MB 71%]
81% [3 Contents-amd64 38.4 MB/53.5 MB 72%]
84% [3 Contents-amd64 41.2 MB/53.5 MB 77%]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [102 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [188 kB]
Fetched 75.1 MB in 15min 32s (80.7 kB/s)
496 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

13.2. Install Sublist3r

sudo apt install sublist3r -y

Install the Sublist3r tool from the Kali Linux repository.

```
┌──(riya㉿kali)-[~]
└─$ sudo apt install sublist3r -y
[sudo] password for riya:
The following package was automatically installed and is no longer required:
  iucode-tool
Use 'sudo apt autoremove' to remove it.

Installing:
  sublist3r

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 496
  Download size: 621 kB
  Space needed: 1,947 kB / 7,914 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 sublist3r all 1.1-5 [621 kB]
Fetched 621 kB in 5s (136 kB/s)
Selecting previously unselected package sublist3r.
(Reading database… 424769 files and directories currently installed.)
Preparing to unpack …/sublist3r_1.1-5_all.deb…
Unpacking sublist3r (1.1-5)…
Setting up sublist3r (1.1-5)…
Processing triggers for kali-menu (2026.2.6)…
Processing triggers for man-db (2.13.1-1)…
```

13.3. Verify the Installation

sublist3r --help

Check whether Sublist3r has been installed successfully.

```
┌──(riya㉿kali)-[~]
└─$ sublist3r --help
usage: sublist3r [-h] -d DOMAIN [-b [BRUTEFORCE]] [-p PORTS] [-v [VERBOSE]] [-t THREADS] [-e ENGINES] [-o OUTPUT] [-n]

OPTIONS:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN   Domain name to enumerate it's subdomains
  -b, --bruteforce [BRUTEFORCE]
                        Enable the subbrute bruteforce module
  -p, --ports PORTS     Scan the found subdomains against specified tcp ports
  -v, --verbose [VERBOSE]
                        Enable Verbosity and display results in realtime
  -t, --threads THREADS
                        Number of threads to use for subbrute bruteforce
  -e, --engines ENGINES
                        Specify a comma-separated list of search engines
  -o, --output OUTPUT   Save the results to text file
  -n, --no-color        Output without color

Example: python3 /usr/bin/sublist3r -d google.com
```

13.4. Basic Syntax

sublist3r -d <domain>

Use the following command to discover subdomains of a target domain.

13.5. Example

sublist3r -d microsoft.com

Discover subdomains for **microsoft.com**.

```
┌──(riya㉿kali)-[~]
└─$ sublist3r -d microsoft.com

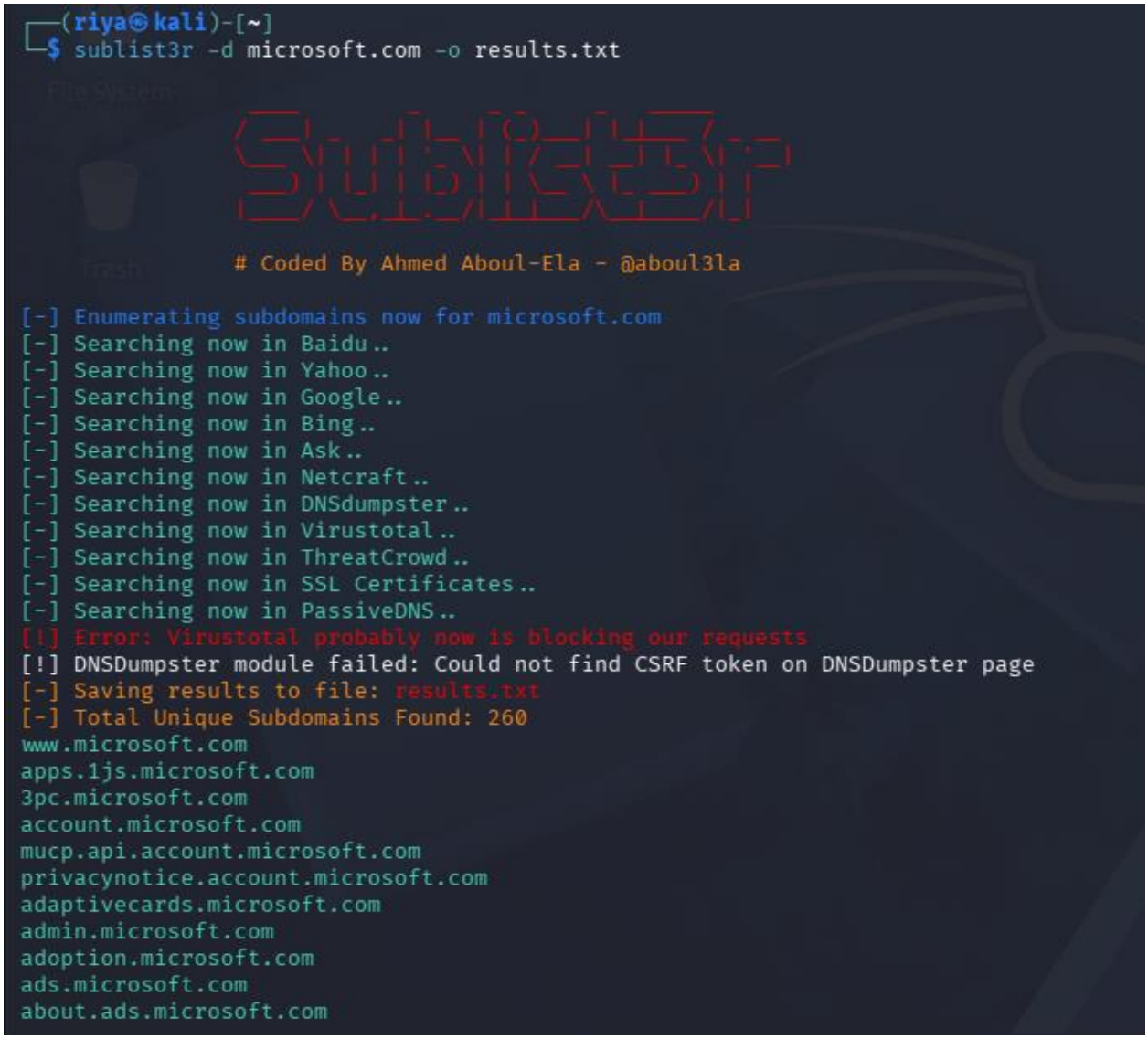
                # Coded By Ahmed Aboul-Ela - @aboul3la

[-] Enumerating subdomains now for microsoft.com
[-] Searching now in Baidu..
[-] Searching now in Yahoo..
[-] Searching now in Google..
[-] Searching now in Bing..
[-] Searching now in Ask..
[-] Searching now in Netcraft..
[-] Searching now in DNSdumpster..
[-] Searching now in Virustotal..
[-] Searching now in ThreatCrowd..
[-] Searching now in SSL Certificates..
[-] Searching now in PassiveDNS..
[!] Error: Virustotal probably now is blocking our requests
[!] DNSDumpster module failed: Could not find CSRF token on DNSDumpster page
[-] Total Unique Subdomains Found: 240
www.microsoft.com
account.microsoft.com
mucp.api.account.microsoft.com
privacynotice.account.microsoft.com
admin.microsoft.com
adoption.microsoft.com
ads.microsoft.com
about.ads.microsoft.com
help.ads.microsoft.com
internal.ads.microsoft.com
trinity.ads.microsoft.com
ui.ads.microsoft.com
aiskillsnavigator.microsoft.com
cdn.aiskillsnavigator.microsoft.com
aitour.microsoft.com
```

13.6. Save the Results

sublist3r -d microsoft.com -o results.txt

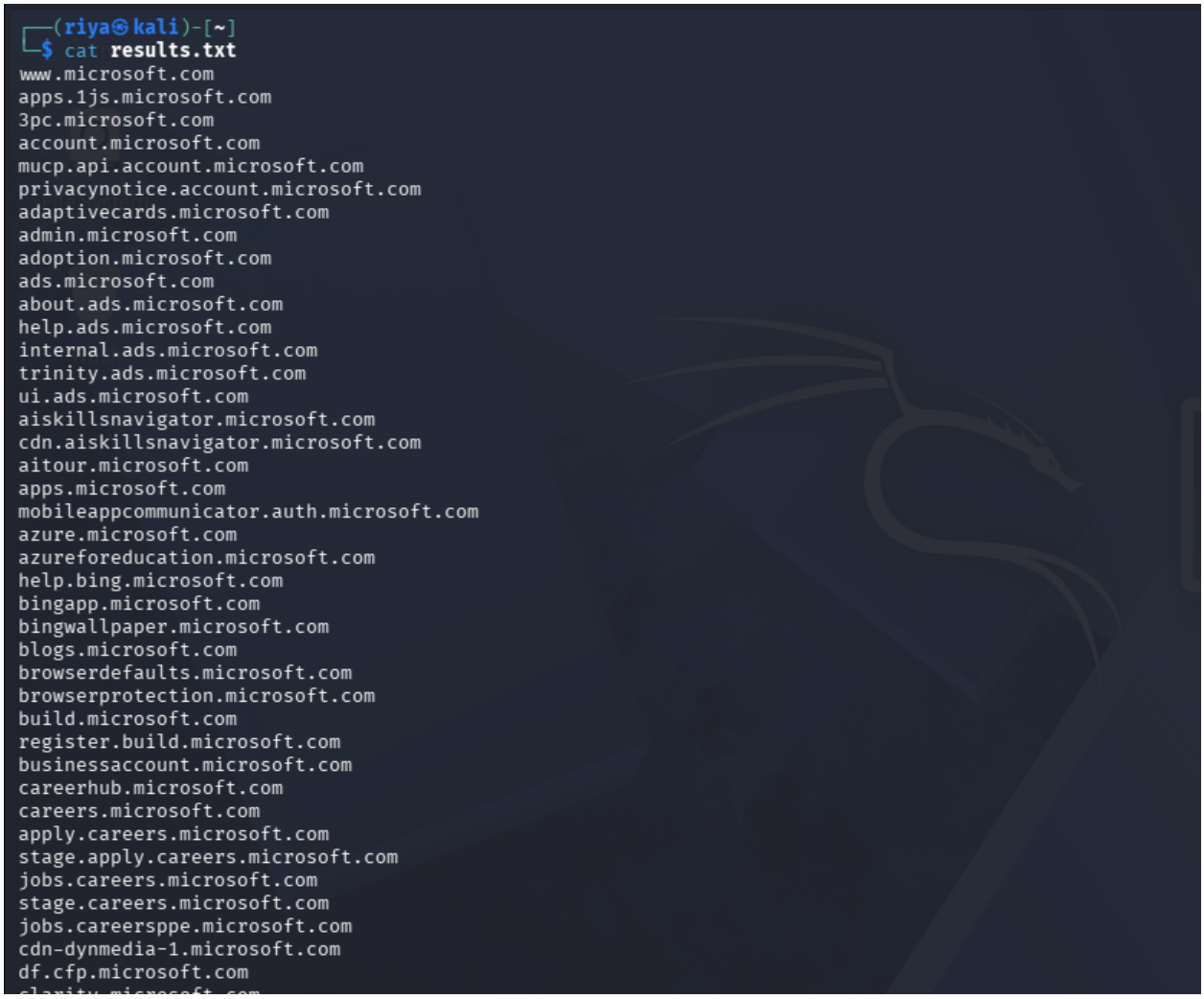
Save the discovered subdomains to a text file.



13.7. View the Saved Results

cat results.txt

Display the contents of the saved output file.


(riya㉿kali)-[~]
$ cat results.txt
www.microsoft.com
apps.1js.microsoft.com
3pc.microsoft.com
account.microsoft.com
mucp.api.account.microsoft.com
privacynotice.account.microsoft.com
adaptivecards.microsoft.com
admin.microsoft.com
adoption.microsoft.com
ads.microsoft.com
about.ads.microsoft.com
help.ads.microsoft.com
internal.ads.microsoft.com
trinity.ads.microsoft.com
ui.ads.microsoft.com
aiskillsnavigator.microsoft.com
cdn.aiskillsnavigator.microsoft.com
aitour.microsoft.com
apps.microsoft.com
mobileappcommunicator.auth.microsoft.com
azure.microsoft.com
azureforeducation.microsoft.com
help.bing.microsoft.com
bingapp.microsoft.com
bingwallpaper.microsoft.com
blogs.microsoft.com
browserdefaults.microsoft.com
browserprotection.microsoft.com
build.microsoft.com
register.build.microsoft.com
businessaccount.microsoft.com
careerhub.microsoft.com
careers.microsoft.com
apply.careers.microsoft.com
stage.apply.careers.microsoft.com
jobs.careers.microsoft.com
stage.careers.microsoft.com
jobs.careersppe.microsoft.com
cdn-dynmedia-1.microsoft.com
df.cfp.microsoft.com